KINCAID IQ · V4 CONSTITUTION · MODULE COMPENDIUM

The Architecture of Receipts.

Every module of the platform — purpose, mechanism, fiduciary consequence — in one place. The 170-module map of how messy reality becomes board-defensible truth.

22	160	3	278+
OPERATING SYSTEMS	DOCUMENTED MODULES	ARCHITECTURAL LAYERS	KPIS COMPUTED

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SIRIUSB IQ · KINCAID IQ · KINCAID RMC
CARMEL, INDIANA · V10.4

CONSTITUTIONAL DOCTRINE · FOUR HARD RULES

One screen, four tiles always. Every number clickable into receipts. No export without receipts. Narratives never outrun evidence.

Module Inventory

22 OPERATING SYSTEMS · 160 DOCUMENTED MODULES · 3 LAYERS

OS ID	NAME	LAYER	MODULES
OS-01	Evidence & Truth	L1	10
OS-02	Ingestion & Indexing	L1	19
OS-03	Workflow & Operations	L1	10
OS-04	Health Plan Intelligence	L1	13
OS-05	Pharmacy / PBM Intelligence	L1	8
OS-06	Actuarial Standards	L1	6
OS-07	Fiduciary & Governance	L1	6
OS-08	Executive Decision	L1	8
OS-09	Security & Access	L1	9
OS-10	Integrations & Export	L1	8
OS-11	Growth & GTM	L1	7
OS-12	Platform Engineering	L1	8
OS-13	Continuous Verification	L2	6
OS-14	Contract Enforcement	L2	5
OS-15	Enterprise Value	L2	4
OS-16	Market Intelligence Network	L2	4
OS-17	Autonomous Governance	L2	4
OS-18	Proof & Attestation	L3	5

OS ID	NAME	LAYER	MODULES
OS-19	Federated Benchmark	L3	5
OS-20	Policy-as-Code	L3	5
OS-21	Ecosystem & Developer Platform	L3	5
OS-22	Remediation Automation	L3	5

OS-01 → OS-12 · 111 MODULES

Layer I · Core Intelligence

The computational substrate. Where messy reality becomes deterministic, hash-stamped, board-defensible fact.

OS-01 10 MODULES

Evidence & Truth

The substrate of fiduciary defensibility. Nothing reaches a board surface without passing through here.

M001 **EvidenceVault**

Immutable, content-addressed object store.

S3-class blob storage where every uploaded artifact — contract PDFs, claims CSVs, DOL 5500 filings, invoices — is sealed under its SHA-256 content hash. Once written, the object is bit-identical for life; any change yields a new hash and a new artifact. This is the foundation of tamper-evident analytics: you cannot rewrite history because history is addressed by its own fingerprint.

M002 **ArtifactRegistry**

Catalog of every artifact the platform has ever seen.

The card catalog of the Vault. Tracks artifact_id, content_hash, mime type, source provenance (uploaded_by, received_at, channel), sensitivity tags (PII / PHI / attorney-client / trade-secret), retention class, and lifecycle state. The Registry is the index layer that makes the Vault searchable and auditable without ever touching the raw bytes.

M003 **HashingAndVersioning**

FIPS 180-4 hashing pipeline and version semantics.

Computes SHA-256 over object content, computes derived hashes (input_hash, evidence_set_hash, transform_hash), and assigns semantic versions to KPIs and transforms. Same inputs → same hash → same output is provable, not asserted. This is what makes the platform *deterministic* rather than merely *automated*.

M004 **LineageGraph**

Directed acyclic graph of dependencies from evidence to claim.

For every published number, the Lineage Graph holds the full DAG: which raw artifacts fed which transforms, which transforms produced which intermediate facts, which facts composed the final KPI. Auditors do not ask 'where did this come from?' — they traverse it. ERISA §404(a)(1)(B) defensibility lives here.

M005 **TransformRegistry**

Versioned catalog of every computation function in the platform.

A transform is code: a normalization routine, a PEPM calculation, a rebate reconciliation. Each is registered with transform_id, code_version (git SHA), params_hash, and a frozen signature. Recomputing a 2023 KPI in 2026 produces the *exact* number it produced then — because the 2023 transform_hash is still on disk and still callable.

M006 **ConfidenceScoring**

Tiering engine: CERTIFIED, MODELED, INSUFFICIENT EVIDENCE.

Every finding is stamped with one of three states. CERTIFIED = direct evidence on file, every input traceable to a specific record. MODELED = actuarial inference with disclosed bounds. INSUFFICIENT EVIDENCE = data missing — and treated as a forensic signal, not a gap. The absence of a number is itself a finding.

M007 **AuditPackBuilder**

Generator of board-defensible evidence bundles.

Packages a published finding into a self-contained, hash-chained PDF + evidence ZIP: the claim itself, every supporting artifact, every transform invocation, every confidence tier, every limitation. Designed to the standard of a legal discovery exhibit — counsel can hand it to opposing counsel and let the math do the talking.

M008 **PolicyControlsEngine**

Runtime enforcement of governance rules.

Where PolicyDSLv1 lives. Evaluates 'is this evidence fresh enough?' 'does this user have RBAC clearance?' 'is the cohort large enough for k-anonymity?' before any value is rendered. Returns PolicyResult(allowed, verified_status, freshness_status, violations[]) — the verdict that the War Room obeys.

M009 **CitationAnchorStore**

Stable, durable URIs for every cited number.

Every figure in a board memo or board pack carries an anchor link back to the exact transform invocation, the exact evidence set, and the exact policy verdict that produced it. Click the number, see the receipt. Forever.

M010 **ChangeLogCompiler**

Append-only ledger of every retroactive edit.

When upstream data corrects itself — late claims adjudication, eligibility correction, vendor-issued reissue — the ChangeLog captures before/after JSONB diffs and row-level hashes. Retroactive truth is detectable, bounded, and governable. Boards see what changed, when, and why.

OS-02 19 MODULES

Ingestion & Indexing

The customs office. Messy reality enters; canonical, hash-stamped facts exit.

M001 **UploadGateway**

Multi-channel intake.

HTTPS upload, SFTP drop, email-to-vault, API push. Authenticates the channel, computes content hash, writes idempotently to the Vault. Same object uploaded twice = no-op (same hash). Eliminates the duplicate-import disaster mode that destroys most analytics platforms in year two.

M002 **PDFParser**

Native-PDF and scanned-PDF extraction.

OCR + structural inference for DOL 5500s, Summary Plan Descriptions, PBM contracts, broker compensation disclosures, Schedule A/C filings. Outputs structured JSON with page-level provenance — every extracted value carries (page, bounding_box) so reviewers can verify visually.

M003 **CSVParser**

Robust CSV/TSV reader with schema tolerance.

Handles delimiter inference, encoding sniffing (UTF-8 / Windows-1252 / Latin-1), header detection, type promotion, null-token normalization. Built for the reality that carrier claims extracts arrive in nineteen different dialects.

M004 **XLSXParser**

Excel workbook decomposition.

Walks multi-sheet workbooks, preserves merged-cell semantics, extracts formulas separately from values, captures defined names and named ranges. Distinguishes input cells from computed cells — critical when a vendor's 'rebate report' is built on hidden Excel logic.

M005 **JSONParser**

API-payload normalizer.

Schema-aware JSON intake for HRIS / eligibility / claims feeds. Validates against registered contracts and rejects malformed payloads at the boundary rather than letting them poison downstream KPIs.

M006 **SchemaInference**

Automated structural detection.

For unknown files, runs column-type inference, key detection, foreign-key suggestion, and confidence scoring on field semantics ('this looks like an NDC,' 'this looks like a member_id'). Surfaces a proposed schema to a human reviewer before promoting to canonical.

M007 **CanonicalNormalizer**

Vendor-dialect → canonical model translator.

Maps the 19 carrier dialects, 14 PBM dialects, and 8 TPA dialects into the canonical claim, eligibility, vendor, and contract schemas. This is where information asymmetry begins to collapse: the same NDC, the same member, the same dollar means the same thing across vendors.

M008 **EntityResolverEmployer**

Probabilistic employer identification.

Resolves 'ACME Corp,' 'Acme Corporation,' 'Acme Corp Inc.,' and EIN 12-3456789 to a single canonical employer_id. Uses Jaro-Winkler on name + EIN cross-check + DOL Form 5500 cross-reference.

M009 **EntityResolverVendor**

Carrier / PBM / TPA / broker resolution.

Same logic for vendors. 'Express Scripts,' 'ESI,' 'Evernorth Care Solutions' collapse to a single vendor_id with full alias history. Required for SBI™ (Shady Broker Index) calculation.

M010 **EntityResolverPlan**

Plan-year-level resolution.

Resolves which plan year a claim or document belongs to under the *employer's* plan calendar (which is not always January–December). Critical for accurate PEPM denominators.

M011 **MultiYearLinker**

Longitudinal cohort assembly.

Stitches plan years together to build 3-, 5-, and 7-year trend lines. Enforces continuity rules: a member must be present in eligibility before claims count, and a vendor change mid-year produces explicit transition flags.

M012 **DriftDetector**

Statistical surveillance over canonical tables.

Watches for distributional shifts — sudden spike in average PMPM, sudden collapse in claim count, sudden change in NDC mix. Drift events feed OS-13 Continuous Verification and trigger evidence-integrity alerts before bad numbers reach the War Room.

M013 **IndexPublisher**

Read-side materialization.

Builds and refreshes the search indices, materialized views, and OLAP cubes the War Room and Sales OS query at runtime. Idempotent; every refresh is hash-tagged so freshness is provable.

M014 **FileTypeDetector**

MIME-aware routing.

Inspects magic bytes (not file extensions) to route each upload to the correct parser. Prevents a renamed .docx from being mistaken for a CSV.

M015 **EncodingNormalizer**

Character-set canonicalization.

Forces every text artifact to UTF-8 NFC. Eliminates the silent corruption mode where 'Müller' becomes 'M??ller' between systems.

M016 **DeduplicationEngine**

Cross-source duplicate suppression.

Detects when the same claim arrives via two channels (carrier feed + employer self-upload). Keeps both copies in the Vault but deduplicates at the canonical-claim layer with full lineage to both sources.

M017 **ChunkAndEmbed**

Document chunking for retrieval.

For contracts and SPDs, chunks at clause boundaries, generates embeddings, and writes them to the vector store that OS-14 Contract Enforcement queries. Always paired with original PDF citations — no answer without a page anchor.

M018 **BatchScheduler**

Bulk-import orchestration.

Coordinates large historical loads (e.g., five years of monthly carrier extracts) with checkpointing, retry, and partial-failure isolation. A failed file fails alone; the batch continues.

M019 **ProvenanceStamper**

Source-attribution writer.

Stamps every canonical row with (source_artifact_id, source_row_offset, ingested_at, ingestion_run_id). This is what makes the LineageGraph traversable: every fact knows its parents.

OS-03 10 MODULES

Workflow & Operations

The factory floor. Where cases move from intake to publication under explicit state and explicit SLA.

M001 **CaseManager**

The unit of analytic work.

A Case is an engagement: a specific employer, a specific scope, a specific clock. CaseManager handles creation, ownership, assignment, status, and closure. Every artifact and every KPI in the system carries a case_id.

M002 **StateMachine**

Enforces the artifact lifecycle.

CREATED → INTAKE_PENDING → PARSED → NORMALIZED → COMPUTED → REVIEW_REQUIRED → PUBLISHED → VERIFIED → ARCHIVED → SUPERSEDED. Illegal transitions are rejected at the database level. No artifact reaches PUBLISHED without verified math or a logged override.

M003 **TaskRouter**

Skill-based work distribution.

Routes review tasks to humans by role, expertise, language, conflict-of-interest constraint, and current load. Actuarial reviews go to actuaries; PHI-bearing reviews go only to PHI-cleared reviewers.

M004 **QueueManager**

Backpressure and prioritization.

FIFO with priority lanes (board-week cases, regulatory deadlines, M&A diligence on closing clock). Visible to operators in real time so nothing rots silently in a queue.

M005 **ApprovalConsole**

Human-in-the-loop publication gate.

The console where reviewers confirm or override the platform's findings before publication. Every decision is logged with reviewer_id, timestamp, rationale, and decision hash — feeding the Decision Memory ledger.

M006 **ExceptionHandler**

Structured error capture.

When a transform fails, the failure becomes an evidence artifact in its own right: which input, which transform_version, which exception class, which stack frame. Failures are not hidden; they are catalogued.

M007 **ManualOverrideManager**

Governed override with explicit cost.

When a reviewer must override an automated finding, the override is logged with rationale, expected duration, sign-off, and an automatic re-review date. Overrides cannot be silent and cannot be permanent without renewal.

M008 **SLAEngine**

Time-based promise enforcement.

Tracks 'evidence bundle generated in ≤15 minutes,' 'board pack delivered in ≤72 hours,' 'vendor packet response in ≤21 days.' Misses generate alerts before they become incidents.

M009 **Notifications**

Multi-channel reviewer signaling.

Email + Slack + in-app. Routed to current owner, escalated on SLA breach, suppressed when redundant. Designed not to be ignored.

M010 **OpsTelemetryEmitter**

Operational metrics for the engineering team.

Throughput, latency, error rate, queue depth — emitted to Prometheus / OpenTelemetry. The platform's own vitals are themselves observable evidence.

OS-04 13 MODULES

Health Plan Intelligence

The fiduciary microscope on the medical plan. Where 'we spent a lot on health' becomes 'we spent $X PEPM, $Y of which is leakage, $Z of which is preventable.'

M001 **PlanIdentitySummary**

Canonical plan-of-record manifest.

Plan year, funding type (fully insured / level-funded / self-funded), carrier, TPA, network, stop-loss carrier and attachment points, ASO fee, plan design tier. The 'face sheet' every other module references.

M002 **ParticipationTrend**

Authoritative member-month rollup.

Computes member-months from eligibility data using half-open interval arithmetic with overlap caps. PEPM denominators are provably correct rather than estimated — the foundation of the Hardening Framework v2.

M003 **FinancialsPEPMPEPY**

Per-employee-per-month and per-year economics.

PEPM = H ÷ (M × P) where H = total spend, M = members, P = plan months. The anchor variable for every volatility model in the platform. Computed deterministically, recomputable to the penny five years later.

M004 **TrendDecomposition**

Year-over-year change attribution.

Decomposes ΔPEPM into utilization Δ, unit-price Δ, mix Δ, and one-time-event Δ. Replaces 'trend talk' with mathematics: when a CFO asks 'why did it go up 11%?', the answer is a waterfall, not a shrug.

M005 **ClaimsDriverAttribution**

Top-claimant and top-condition decomposition.

Where the dollars actually went: top 1% / 5% / 10% claimants, top 10 conditions, top 25 NDCs, site of service mix. The contour map of risk.

M006 **UtilizationPatterns**

Service-line utilization analytics.

ER visits per 1,000, inpatient admits per 1,000, specialist visit ratio, advanced imaging rate, ambulatory surgery vs hospital outpatient mix. The behavioral signal beneath the financial outcome.

M007 **HighCostConcentration**

Catastrophic claimant risk geometry.

Distribution of plan spend across claimant deciles. Stop-loss adequacy testing. Identification of emerging catastrophic claimants (e.g., new oncology, new gene therapy) before they breach the attachment point.

M008 **NetworkOONExposure**

Out-of-network leakage quantification.

Dollars and visits flowing to non-network providers, balance-billing exposure, surprise-billing-act applicability, and the dollar value of network-steering interventions.

M009 **SiteOfCareOpportunities**

Setting-of-service optimization.

Identifies infusions, imaging, and surgical procedures performed at hospital-outpatient sites that are clinically appropriate at ambulatory or freestanding facilities. Quantifies the savings opportunity in dollars and Notre-Dame-grade member friction.

M010 **VendorFeeLedger**

Administrative fee transparency.

ASO fees, network access fees, care management fees, utilization review fees, broker commissions, and the spread between disclosed and actual. The end of 'what are we actually paying for, exactly?'

M011 **StopLossRiskSignals**

Reinsurance position monitoring.

Attachment point adequacy under Monte Carlo with GLP-1, gene therapy, and catastrophic-event scenarios. Lasers, aggregating specifics, and the actuarial honesty of saying when a contract is structurally underpriced.

M012 **ComplianceSignals5500**

DOL Form 5500 cross-reference.

Joins live plan data to the 757K-row 5500 database (64 files, 11 schedule types). Flags inconsistencies between what is filed federally and what is happening operationally — the kind of mismatch that becomes a DOL audit finding.

M013 **BenchmarkIndexing**

Cohort-relative position.

Computes the employer's PEPM, trend, and structural metrics against industry / size / geography cohorts using OS-19's federated benchmarking. Tells the board where it sits — quartile, decile, peer rank.

OS-05 8 MODULES

Pharmacy / PBM Intelligence

The platform's sharpest edge. Where PBM opacity meets deterministic reconciliation.

M001 **RxSpendDecomposition**

Drug-spend anatomy.

Brand vs generic vs specialty, retail vs mail vs specialty channel, top therapeutic classes, top NDCs, member concentration. The first table every pharmacy review begins with.

M002 **SpecialtyDrivers**

Specialty-tier risk model.

Specialty spend trajectory, new-start velocity, autoimmune / oncology / rare-disease mix, biosimilar penetration. Specialty is where most plans lose the next dollar; this module locates it.

M003 **ChannelSteeringSignals**

Distribution-channel economics.

When the PBM steers a specialty drug to its own affiliated specialty pharmacy at non-market pricing, this module finds it. Compares ingredient cost across channels for identical NDCs and quantifies the steering tax.

M004 **PricingIntegrityChecks**

AWP / WAC / NADAC / MAC reconciliation.

For every claim: was the contractually-promised pricing methodology actually applied? Was the discount delivered? Was the dispensing fee correct? The forensic engine behind every Rx Defense finding.

M005 **RebateOpacityScoring**

Manufacturer-rebate transparency analysis.

Tracks contractual rebate guarantees vs realized passthrough. Class-level inference where direct evidence is missing. Quantifies under-remittance as (guaranteed_pct × eligible_spend) − received_dollars. This is where most of the recoverable EBITDA lives.

M006 **FormularyIncentiveMap**

Tier-design conflict surfacing.

Identifies cases where formulary placement appears driven by rebate economics rather than clinical or net-cost optimization. The classic example: high-rebate brand preferred over lower-net-cost generic or biosimilar.

M007 **ContractClauseIntelligence**

PBM agreement decomposition.

Extracts every financially-material clause — spread definitions, audit rights windows, MAC governance, pricing exhibits, performance guarantees — and binds each to runtime reconciliation tests in OS-14.

M008 **RxInterventionScenarios**

Recoverable-dollar modeling.

What does the savings curve look like if we move from spread to passthrough? From three-tier to four-tier? From open to closed formulary? Modeled with confidence bounds and member-disruption forecasts — Notre Dame undercurrent enforced.

OS-06 6 MODULES

Actuarial Standards

The platform's professional conscience. Where Society of Actuaries discipline lives as code.

M001 **KPIStandardsLibrary**

Definitional registry for every KPI.

Each of the 278+ KPIs has a unique definition, a formula in mathematical notation, a code implementation, an evidence requirement, and a version. No KPI can ship without a registered definition — eliminating the 'what does PEPM mean here?' ambiguity that destroys most benefits analytics.

M002 **ContinuityChecks**

Period-over-period sanity verification.

Validates that a 7-year trend line is internally coherent: no missing months, no double-counted months, no eligibility step-functions without explanatory events. Surfaces discontinuities before they distort visuals.

M003 **DataSufficiencyScoring**

Pre-publication evidence audit.

For every KPI computation, scores whether the underlying evidence set meets the minimum-evidence requirement in OS-01's PolicyControlsEngine. Insufficient → INSUFFICIENT EVIDENCE tier; never published as CERTIFIED.

M004 **AttributionMathPack**

Drivers-of-change decomposition library.

The mathematical primitives behind TrendDecomposition: utilization-vs-price, mix-shift, large-claim, network-shift. Implemented as reproducible recipes so attribution math is always the same math, never the analyst's improvisation.

M005 **AssumptionsRegister**

Explicit, signed-off assumption log.

When a forecast requires assumptions (trend rate, GLP-1 utilization growth, stop-loss multiplier), each is recorded with source, value, rationale, owner, and sign-off date. No assumption is anonymous; no assumption is unchallenged.

M006 **LimitationsGenerator**

Mandatory limitations-section author.

Every report ships with a limitations section. This module composes it automatically from the assumption register, the data sufficiency score, and the confidence tiers used. Honest output is the only output.

OS-07 6 MODULES

Fiduciary & Governance

The ERISA spine. Where 'fiduciary duty' stops being a phrase and becomes an inspectable artifact.

M001 **FiduciaryRiskRegister**

Catalog of plan-level fiduciary exposures.

Each entry: risk description, ERISA section reference (§404, §406, §408), evidence on file, current posture, remediation owner, due date. The document a DOL investigator would ask to see first — and the document plan sponsors should review quarterly.

M002 **DisclosureTracker**

ERISA §408(b)(2) compliance ledger.

Tracks broker, consultant, and vendor compensation disclosures. Flags missing, late, or materially inconsistent disclosures. ERISA does not require disclosure to be helpful; it requires it to be on file. We track both.

M003 **ConflictIncentiveMapper**

Compensation-flow graph.

Maps every dollar of vendor/broker/consultant compensation to its source, including indirect (override, contingent, soft-dollar) compensation. Visualizes who is incentivized to recommend what — the structural input to SBI™.

M004 **GovernanceControlsDashboard**

Plan-committee meeting infrastructure.

Agendas, minutes, action items, decision memory, and document retention for benefit committees. The 'we governed it' evidence package that turns a process into a defense.

M005 **LitigationReadyReporting**

Pre-formatted exhibit generation.

When the platform's findings need to become an exhibit — discovery response, regulator inquiry, M&A representation — this module produces the package: claim, evidence, lineage, transform receipt, sign-off chain. Print, seal, deliver.

M006 **ProcurementSignals**

RFP-readiness scoring.

When the data shows the incumbent vendor is structurally underperforming, this module composes the RFP signal: what to ask, what to require, what to disqualify, and what the credible savings range is. Fiduciary process, productized.

OS-08 8 MODULES

Executive Decision

The CFO War Room. Where healthcare stops being an HR topic and starts being a board topic.

M001 **EBITDA_DragModel**

Quantification of current EBITDA erosion.

ΔE = −(H × t). Every 1% healthcare trend at scale collapses to a specific dollar of EBITDA drag. This module computes it from live financials: H = current healthcare spend, t = trend rate, output = annualized EBITDA hit.

M002 **EBITDA_RecoveryModel**

Quantification of recoverable EBITDA.

For each finding from OS-04 and OS-05 — leakage, misalignment, contractual underperformance — the recoverable EBITDA model computes the dollar opportunity, the time-to-capture, and the realization confidence. The line item the CFO actually wants to see.

M003 **ScenarioEngine**

Base / Adverse / Stress projection.

Forward modeling under three named scenarios. Renewal shock, GLP-1 surge, catastrophic claimant, stop-loss restructure. 10,000-iteration Monte Carlo with Cholesky correlation. Probability-weighted EBITDA distribution bands.

M004 **InitiativePortfolio**

Recoverable-opportunity inventory.

Every identified savings opportunity becomes an Initiative: dollar value, owner, due date, dependencies, evidence threshold, status. A board sees the portfolio, not a memo.

M005 **RealizationTracking**

Did we actually capture the dollars?

Closes the loop between projected and realized. Six months after an initiative shipped, this module reconciles forecast vs reality and feeds the variance back into the confidence-scoring engine. The platform learns from its own forecasts.

M006 **KPICommandCenter**

The four-tile War Room.

One screen, four tiles always. Trend / Driver / Risk / Action. Every number clickable into receipts. The interface that enforces the Four Hard Rules architecturally.

M007 **BoardMemoGenerator**

Director-grade narrative composition.

Composes the board memo from the certified facts: executive summary, financial reconstruction, leakage analysis, contract performance, actionable strategy. Narratives never outrun evidence; every assertion is anchored.

M008 **ExecBriefGenerator**

CFO-grade short-form synthesis.

Two pages, four tiles, ten numbers, one recommendation. For the CFO who has fifteen minutes before the next call.

OS-09 9 MODULES

Security & Access

Defense in depth. PHI does not leak; PII does not wander; tenants do not see each other's data.

M001 **IdentityProviderAdapter**

Federated SSO across identity providers.

Okta, Entra ID, Google Workspace, Ping. SAML 2.0 + OIDC. MFA enforced; passwordless preferred. The identity layer is never the platform's; it is always the customer's.

M002 **RBACEngine**

Role-based access with nine canonical roles.

Admin, Reviewer, Auditor, Analyst, Executive, Counsel, Vendor, External Auditor, Read-Only. 42 permissions across the matrix. Permissions are JWT-claimed at API boundary and re-checked at data layer.

M003 **TenantIsolationGuard**

Cross-tenant leakage prevention.

Every query carries tenant_id as a runtime PostgreSQL session variable. Row-Level Security policies enforce it at the database. Integration tests assert isolation continuously. Defense in depth across three layers.

M004 **RowLevelSecurityRules**

Database-enforced data boundaries.

PostgreSQL RLS policies on every canonical table. A query that omits tenant_id returns zero rows by construction — not by convention. The 'forgot the WHERE clause' failure mode is impossible.

M005 **SecretsAndKeys**

Cryptographic material management.

Vault-backed secrets, automatic rotation, KMS-managed encryption keys for data-at-rest. No secret ever in code; no secret ever in a config file.

M006 **AuditLogging**

Immutable access ledger.

Every read, write, export, and override is logged with (user_id, tenant_id, resource_id, action, timestamp, ip, user_agent). Append-only. Hash-chained. The artifact that gets subpoenaed.

M007 **ExportControls**

PHI/PII redaction at export boundary.

Outbound data is filtered against the recipient's clearance, the artifact's sensitivity_tags, and the export class. PHI cannot leave to a non-PHI-cleared recipient by accident.

M008 **RetentionPolicies**

Lifecycle-managed deletion.

Per-artifact-class retention windows aligned to HIPAA / ERISA / state law. Retention expirations trigger reviewed (not automatic) deletion through the ApprovalConsole.

M009 **RedactionEngine**

Configurable masking.

For partial-disclosure scenarios (board materials with PHI redacted, vendor packets with member identifiers masked), the redaction engine applies deterministic, reversible masking — auditable but private.

OS-10 8 MODULES

Integrations & Export

The semipermeable membrane. Data flows in from operational systems; receipts flow out to boards, banks, and counsel.

M001 **ClaimsConnector**

Carrier and TPA claims-feed integration.

Direct ingestion from Aetna, UHC, Anthem, Cigna, BCBS plans, Meritain, HealthEZ, Trustmark, and ~80% of the self-funded market by member-count. SFTP / API / vendor portal scraping where required.

M002 **EligibilityConnector**

HRIS and benefits-admin integration.

Workday, ADP, UKG, Paycom, BambooHR. Real-time member status, plan election, dependent coverage. The denominator side of every PEPM.

M003 **HRISConnector**

Org-chart and compensation context.

Where required and authorized: division / location / job-family slicing for benefits analytics. Enables 'why is healthcare cost in the Memphis plant 22% higher than Indianapolis?' diagnostic depth.

M004 **ERPConnector**

Finance-system reconciliation.

NetSuite, SAP, Oracle, QuickBooks-Enterprise. Reconciles cash paid vs benefits invoices vs carrier-reported claims. Closes the loop that finance teams normally close manually in spreadsheets at quarter end.

M005 **DataExport**

Outbound CSV / Parquet / API.

Customer-controlled extracts of their own canonical data. Their data, on their terms, with full receipts.

M006 **BoardPackExporter**

Board-pack PDF + evidence ZIP.

Four-tile War Room rendered to PDF + every supporting artifact bundled. SHA-256-sealed; downloadable; reproducible.

M007 **WebhookPublisher**

Event-driven outbound.

On Initiative completion, KPI threshold breach, evidence bundle seal, or board pack publication, emits webhook to customer endpoints. Drives Slack alerts, Salesforce updates, JIRA tickets.

M008 **APIGateway**

Programmatic access surface.

REST + GraphQL for customers who want to embed Kincaid IQ data in their own systems. Rate-limited, scoped, logged.

OS-11 7 MODULES

Growth & GTM

The commercial engine. Built into the platform, not bolted onto it.

M001 **ICPAccountTiering**

Ideal-customer-profile scoring.

Self-funded employers 250–25,000 lives, PE portfolio companies, family-office holdings. Scores prospects on size, funding, PBM, broker structure, and DOL 5500 signal.

M002 **ProspectSignals**

Public-data prospect intelligence.

DOL 5500 changes, broker switches, M&A activity, leadership changes, lawsuit filings. The signal layer beneath outbound.

M003 **ABMPlaybooks**

Account-based outbound choreography.

Per-account sequence: research → POV → meeting → diligence → POC → close. Codified so the GTM team executes process, not improvisation.

M004 **ContentEngineIndex**

Thought-leadership catalog.

Forensic reports, monographs, board briefs, LinkedIn articles. Indexed, versioned, and citable — the marketing arm of the evidence spine.

M005 **POVLibrary**

Point-of-View prospect deliverables.

Pre-deal forensic teasers built from public DOL 5500 data: 'here is what we can see about your plan from the outside; imagine what we see when we see inside.'

M006 **SalesEnablementKit**

AE-grade playbook + objection handler.

Discovery scripts, pricing scenarios, mutual NDA templates, POC scope, board-day choreography. Standardized so every AE pitches the same product.

M007 **FunnelMathDashboard**

Pipeline analytics with discipline.

MQL → SQL → Opportunity → Closed-Won with cohort velocity, win-rate by segment, and CAC payback. The internal mirror of the rigor we hold customers to.

OS-12 8 MODULES

Platform Engineering

The engineering backbone. What makes determinism a daily reality rather than an aspiration.

M001 **DeterministicRunMode**

Bit-identical execution guarantee.

Same inputs + same code + same config = same outputs, byte for byte. Enforced via input_hash + transform_hash + config_hash composite cache key. The architectural pillar of audit reproducibility.

M002 **TestHarnessGoldenFiles**

Frozen-output regression testing.

For every KPI, a golden file captures (input fixture, expected output). Any code change that perturbs a golden output fails CI. Drift is impossible without an explicit, reviewed approval.

M003 **RegressionSuiteKPIs**

Continuous correctness verification.

130+ KPIs in the v3 engine; full suite runs on every commit. Catches the silent-correctness bug — where the test passes but the math changed — that destroys analytics platforms in year three.

M004 **ConfigVersioning**

Configuration as code.

Every customer config (thresholds, attachment points, plan-year boundaries, vendor mappings) is versioned in source control with semantic versioning. Rollback is a git revert; change is a pull request.

M005 **FeatureFlags**

Runtime capability gating.

Feature rollouts staged per tenant, per role, per environment. New KPIs ship to internal review before customers, to design partners before general availability.

M006 **ObservabilityMetrics**

Operational vitals.

Throughput, latency, error rate, queue depth — every module emits Prometheus metrics. Grafana dashboards visualize the platform's own health. Self-observability is non-negotiable.

M007 **RunbookGenerator**

Operational documentation as artifact.

Every service has a generated runbook: dependencies, alerts, escalation, recovery procedures. The on-call engineer is never reading source for the first time at 3 AM.

M008 **PluginFramework**

Extensibility without forking.

Customer-specific transforms (a unique vendor format, a unique compliance check) plug in via the framework without modifying core code. Customization without fragmentation.

OS-13 → OS-17 · 23 MODULES

Layer II · Continuous Verification

Always-on enforcement. Drift surveillance, contract reconciliation, fiduciary scoring — running in real time against every published claim.

OS-13 6 MODULES

Continuous Verification

The standing army. Always on, watching every signal, enforcing publishing law in real time.

M001 **EventIngestionStream**

Kafka-backed event spine.

Every ingestion, every transform, every state change, every export emits a structured event. The verification layer subscribes and acts. Real time, not batch.

M002 **DriftSurveillanceEngine**

Statistical drift detection.

Watches distributions for changes that exceed configured thresholds — population shift, claim-mix shift, vendor-fee shift. Drift events feed alerts and may auto-suspend publication for affected KPIs until reviewed.

M003 **PolicyEnforcementTriggers**

Runtime policy evaluation.

When a published KPI ages past its freshness SLA, when evidence is superseded, when a vendor contract amends — triggers fire and PolicyResults update. Stale truth becomes unverified truth instantly.

M004 **LiveRiskIndex**

Real-time fiduciary-risk scoring per plan.

A continuously-updated 0–100 index per employer combining contract risk, drift risk, evidence-freshness risk, and finding-severity risk. The dial the CFO checks every morning.

M005 **VerificationScheduler**

Periodic re-verification cadence.

Even KPIs that don't drift get re-verified on schedule (monthly / quarterly / annually). Verification is a habit, not an event.

M006 **AlertRouter**

Severity-graded notification delivery.

INFO / WARN / CRITICAL routed to the appropriate audience: ops engineer for system events, plan committee for fiduciary events, board chair for material EBITDA events. Right alert, right audience, right channel.

OS-14 5 MODULES

Contract Enforcement

The contract becomes code. What the document says, the platform enforces.

M001 **ClauseExtractionAndStructuring**

Contract → structured clause objects.

PBM agreements, ASO contracts, stop-loss treaties, broker agreements, network access deals. Every financially-material clause becomes (clause_id, category, language, parameters, risk_tier, dollar_exposure_range).

M002 **ClauseVectorizationToPolicy**

Clause → executable policy.

Each clause compiles into a PolicyDSLv1 rule: 'rebate passthrough ≥ 100% of class X' becomes a runtime test against claims data with a measurable pass/fail per period.

M003 **AuditRightsMonitor**

Audit-window tracking.

Most PBM and ASO contracts contain finite audit-rights windows. This module tracks every window, alerts in advance of expiration, and prepares the audit packet automatically — turning a missed deadline (and millions in foreclosed claims) into a routine task.

M004 **GuaranteeValidator**

Performance-guarantee reconciliation.

Generic dispensing rates, mail-order penetration, customer-service response time, claim-payment accuracy. Each guarantee is reconciled against actual performance every quarter with explicit pass/fail and dollar-credit calculation.

M005 **ContractRiskScoring**

Per-contract risk index.

Combines clause-density, opacity, audit-rights tightness, termination flexibility, and guarantee strength into a 0–100 score. A contract earning 38 is a renewal priority; a contract earning 88 is acceptable.

OS-15 4 MODULES

Enterprise Value

The translation layer. Where benefits cost becomes enterprise value, EBITDA, and capital allocation.

M001 **ValueSensitivityEngine**

ΔEBITDA / ΔEV per intervention dollar.

For every potential remediation, compute the sensitivity: a $1 dollar of recovered Rx leakage at 12× multiple = $12 of EV. The math that makes a benefits decision a finance decision.

M002 **FiduciaryStabilityIndex**

FSI™ — proprietary 0–100 fiduciary-stability score.

Composite of contract risk, evidence freshness, governance completeness, and disclosure quality. The Moody's-style rating that PE deal teams, family offices, and boards ask for during diligence.

M003 **DealDiligencePackGenerator**

M&A-ready data room artifact.

Three-year benefits forensic, EBITDA-impact reconstruction, fiduciary posture, contract-risk summary, FSI™ score, recoverable-EBITDA projection. The package a PE deal team handed in week one is the package that reprices week three.

M004 **CapitalAllocationSignals**

Healthcare-as-capital-allocation framing.

Treats healthcare spend as deployed capital with a return profile. EVA = NOPAT − (Capital × WACC). Where the return is negative, the platform says so. The Stern Stewart discipline imported into benefits.

OS-16 4 MODULES

Market Intelligence Network

Privacy-preserving collective intelligence. The whole network learns; no client's data leaks.

M001 **AnonymizationAndAggregation**

K-anonymity enforcement in code.

Cohort minimum sizes enforced before any benchmark renders. Differential-privacy-lite noise where appropriate, explicitly labeled (never overclaimed). Privacy is a code property, not a policy aspiration.

M002 **BenchmarkEngine**

Federated cohort comparisons.

PEPM, trend, claim-mix, contract-economics distributions across industry / size / geography. The customer sees their position; they never see another customer's data.

M003 **VendorPatternDetection**

Cross-employer vendor-behavior surveillance.

When the same PBM under-remits rebates in similar patterns across multiple employers, the network detects it before any single employer would. Class-action-grade evidence assembled responsibly.

M004 **EmergingRiskRadar**

Forward signal aggregation.

GLP-1 utilization curves, gene-therapy claim counts, new-vendor entry, regulatory shifts. The early-warning system for risks that haven't yet hit the median employer.

OS-17 4 MODULES

Autonomous Governance

The platform's own conscience. Catches what humans miss and routes it back to humans.

M001 **EvidenceGatekeeper**

Runtime enforcement of publishing law.

Before any number renders to a War Room tile, the Gatekeeper confirms: evidence present, evidence fresh, transform verified, policy passed, confidence tier set. Fails closed. No exceptions, ever.

M002 **AutoExceptionLogger**

Automatic capture of every governance breach.

When the Gatekeeper blocks something, when an override fires, when a SLA slips — captured, classified, ticketed automatically. The exception log is itself evidence of governance discipline.

M003 **GovernanceDriftMonitor**

Surveillance of the governance system itself.

Detects when too many overrides are firing, when reviewer approval times are growing, when evidence freshness is degrading across the platform. The platform watches its own discipline and surfaces the slippage.

M004 **DecisionQualityScoring**

Outcome tracking on past decisions.

When an Initiative completes, did the realized recovery match the forecast? When a stop-loss attachment was raised, did the catastrophic-claim curve validate the call? The platform measures its own quality of judgment and learns.

OS-18 → OS-22 · 24 MODULES

Layer III · Fiduciary Control Plane

Cryptographic proof, federated benchmark, policy-as-code, remediation automation. The mathematical end of trust.

OS-18 5 MODULES

Proof & Attestation

Cryptographic, independently verifiable. The mathematical end of trust.

M001 **ProofManifestGenerator**

Signed manifest per published computation.

Per claim: artifact_hashes, transform_ids, config_version, build_id, signing_key_id. A third-party auditor with the manifest and the source code can rebuild the claim and verify the byte-identical output.

M002 **AttestationSigner**

Cryptographic signing of manifests.

Ed25519 signatures binding the manifest to the issuing tenant and platform version. Tampering breaks the signature; the breakage is provable; the proof is portable.

M003 **ProofVerificationAPI**

Third-party reverification endpoint.

Auditor / counsel / regulator endpoint: submit a manifest + a claimed value; receive verified/unverified plus the discrepancy if any. Trust becomes mechanical.

M004 **HashChainLedger**

Append-only ledger of every published claim.

ledger_hash_n = SHA256(prev || input || result). Each link binds to its predecessor. Retroactive insertion is mathematically impossible. The hash chain is the platform's public memory.

M005 **AttestationDashboard**

Customer-facing verification surface.

For every number a customer sees, a 'verify' affordance opens the manifest, shows the receipt, and links to the independent verification endpoint. Trust, externalized.

OS-19 5 MODULES

Federated Benchmark

Network-grade benchmarking without centralization. Privacy-preserving by construction.

M001 **FederatedAggregator**

Multi-tenant statistic aggregation without raw data centralization.

Per-tenant statistics computed locally, aggregated at the network layer under k-anonymity minima and configurable noise. No tenant's raw claims, eligibility, or contracts ever cross the tenant boundary.

M002 **PrivacyBudgetTracker**

Differential-privacy accounting.

Every published benchmark consumes a budget; budgets are tracked per cohort to prevent re-identification through repeated querying. Honest privacy, not theatrical privacy.

M003 **CohortDefinitionRegistry**

Comparable-peer-group definitions.

Industry × size × geography × funding type. Cohorts are defined once, versioned, and reused — so the 'median PEPM' a customer sees today is comparable to the 'median PEPM' they saw a quarter ago.

M004 **BenchmarkPublisher**

Cohort-statistic distribution to customers.

Pushes updated quartile / decile / median statistics to customer surfaces on a schedule, with full transparency about cohort size, freshness, and confidence.

M005 **ParticipationGovernance**

Opt-in / opt-out customer control.

Customers choose whether their (anonymized) statistics contribute to the network.
Participation is opt-in by tenant decision; never assumed; revocable.

OS-20 5 MODULES

Policy-as-Code

Governance lives in source control. Rules are versioned, testable, reviewable, enforceable.

M001 **PolicyDSLv1**

Domain-specific language for governance rules.

Declarative syntax: freshness_max_hours: 72; min_evidence: 3; cohort_min: 30. Compiled at deploy; enforced at runtime. The policy is the code; the code is the policy.

M002 **PolicyRegistry**

Versioned catalog of every active policy.

Per policy: id, version, scope (global / tenant / case), parameters, owner, effective_date, supersedes. Policies follow semantic versioning. A change is a PR.

M003 **PolicyTestHarness**

Unit and integration tests for governance rules.

Before a policy goes live, the harness validates it against the regression suite. A policy that would break a published KPI cannot deploy without explicit acknowledgment.

M004 **PolicyAuditTrail**

Every policy evaluation logged.

(policy_id, version, input, result, timestamp). Auditors can ask 'what policy was in force when this number published, and what did it return?' and receive a one-row answer.

M005 **PolicyAuthoringConsole**

Reviewer-grade UI for policy edits.

Drafts, diffs, simulations, approval workflow. Policy authors do not edit YAML in a vacuum; they author through a console that previews impact across the platform.

OS-21 5 MODULES

Ecosystem & Developer Platform

Kincaid IQ as substrate. The category leader is the one others build on.

M001 **PublicAPIContract**

Versioned, documented external interface.

REST + GraphQL with OpenAPI / GraphQL-schema published. Semantic versioning; deprecation windows; backward compatibility for n−1 minor versions.

M002 **PartnerSDKs**

First-party client libraries.

Python, TypeScript, Java. Idiomatic for each language; auto-generated from the API contract; published to PyPI / npm / Maven Central.

M003 **IntegrationMarketplace**

Catalog of partner-built connectors and extensions.

Listing process, certification standard, revenue share. The ecosystem flywheel for Kincaid IQ-as-platform.

M004 **CustomerDataBoundaryKit**

Tooling for customer-controlled deployments.

BYOK encryption, dedicated tenant infrastructure, on-prem evidence vault for the largest customers who require data-residency guarantees beyond cloud.

M005 **CompliancePortal**

Real-time customer-facing compliance evidence.

SOC 2 Type II reports, HIPAA Business Associate documentation, penetration-test summaries, sub-processor registry. Updated continuously; downloadable on demand.

OS-22 5 MODULES

Remediation Automation

Findings become actions become outcomes. The loop closes inside the platform.

M001 **ContractRedlineGenerator**

Marked-up amendment proposals.

When a clause is underperforming or absent, this module generates the redline: the exact insertion / deletion / substitution language, tied to the finding that justified it. Customer counsel reviews; the platform did the legal scaffolding.

M002 **RFPClauseGenerator**

RFP-ready clause libraries.

When the data justifies a market test, the platform produces the RFP package: required clauses, scoring rubric, vendor disqualification criteria, model contract terms. The procurement playbook, productized.

M003 **DisputePacketBuilder**

Vendor-dispute evidence assembly.

When reconciliation finds a discrepancy, this module assembles the dispute packet: claim, evidence, computation, contractual basis, requested remedy. Send-ready; counsel-reviewed; auditor-defensible.

M004 **InitiativeExecutor**

Workflow-driven remediation orchestration.

An Initiative is a project. The Executor manages task assignment, dependency tracking, milestone progression, and realization reconciliation — closing the loop from finding to recovered dollars.

M005 **BoardActionEngine**

Board-decision capture and tracking.

When a board approves a remediation path, the engine records the resolution, links it to the Initiative, tracks execution, and reports back at the next meeting. Decision Memory, materialized.

Dashboards show numbers.
Kincaid IQ shows receipts.

SIRIUSB IQ · AI FINANCIAL INTELLIGENCE INFRASTRUCTURE